

AuditIQ Pro

AI-Powered Assurance Intelligence

Development & Technical Report

1. Executive Summary & Project Context

- **Project Title:** AuditIQ Pro
- **Tagline:** AI-Powered Assurance Intelligence
- **Purpose:** The project aims to reduce repetitive audit work, improve efficiency, support risk assessment, streamline planning activities, and centralize audit workflows in a single platform. Furthermore, with additional features of creation of planning memo, management letter with advanced features of AI audit copilot and knowledge hub.
- **Context:** Developed as part of an AI-assisted software development to explore how modern AI tools can accelerate application development while solving a real-world business problem from doing Traditional Audit to AI Powered Audit & Assurance.

2. Problem Statement & Solution

The Real-World Problem

Auditors spend significant time on documentation, planning, risk assessment, and information retrieval. These repetitive and manual processes can create inefficiencies throughout the audit lifecycle.

The Solution (App Idea)

AuditIQ Pro provides a centralized workspace where auditors can manage engagements, assess risks, analyze financial data, generate audit

documentation, maintain audit checklists, and interact with an AI-powered audit copilot.

Does the Platform Solve a Real-World Problem?

The platform addresses practical audit challenges including documentation effort, planning inefficiencies, knowledge retrieval, and risk assessment support. This platform is built based on solving the real-world problems and difficulties faced by the auditors during audit engagements / assignments.

3. Technology Stack & AI Tooling

The application leverages a modern full-stack architecture integrated with high-performance AI models:

Technology Stack Table

Layer	Technologies Used
Frontend	React, Vite, Tailwind CSS, React Router
UI/UX	Lenis, GSAP, Three.js, Lucide Icons
Backend	Supabase Authentication, PostgreSQL Database, Row Level Security, Edge Functions
AI Layer	Groq API
Deployment	Vercel

AI Tools Utilized During Development

- **ChatGPT:** Used for architecture planning, prompt engineering, debugging, feature design, and documentation.
- **VS Code / AI Coding Assistants:** Used for code and refactoring.
- **Groq API:** Used to power AI-driven features such as working papers, procedures, planning memos, management letters, audit copilot responses, and knowledge hub functionality.

4. AI-Assisted Development & Prompt Engineering

The Development Lifecycle

The application followed a continuous loop of AI-assisted iteration: Idea -> Prompt -> Generated Output -> Testing -> Issue Identification -> Prompt Refinement -> Improved Output -> Repeat

Initial Prompt Used

The development process began with the assignment objective of building an AI-assisted application that solves a real-world problem. The idea evolved into an AI-powered audit copilot because of the strong alignment with audit productivity challenges.

Prompt Version: 1

Goal: Generate first static version of AuditIQ Pro.

Input Given: App idea, modules, UI requirements, folder structure.

Output Received: React + Tailwind dashboard with placeholder screens.

Problems Found: To be filled after testing.

Improvements Needed: Supabase, Gemini, financial analysis, charts, real logic.

Result: First MVP skeleton created.

Prompt Version: 2

Goal: Add risk scoring logic.

Input Given: Risk factors and scoring levels.

Output Received: Interactive risk assessment module.

Problems Found: To be filled after testing.

Improvements Needed: Financial statement analyzer and charts.

Result: Risk engine added successfully.

Prompt Version: 3

Goal: Add financial statement analyzer.

Input Given: CSV format, ratio formulas, red flag rules, chart requirements.

Output Received: CSV upload, ratio cards, financial charts, red flag detection.

Problems Found: To be filled after testing.

Improvements Needed: Supabase database and Gemini AI generation.

Result: Financial analyzer added successfully.

Prompt Version: 4

Goal: Add Supabase backend integration.

Input Given: Database tables, environment variables, save logic, error handling.

Output Received: Supabase client, project saving, risk assessment saving.

Problems Found: To be filled after testing.

Improvements Needed: Gemini AI integration through Supabase Edge Function.

Result: Backend storage added successfully.

Prompt Version: 5

Goal: Add Gemini API through Supabase Edge Function.

Input Given: AI modes, function structure, security requirement, output formats.

Output Received: Gemini-powered working papers, procedures, and audit copilot.

Problems Found: To be filled after testing.

Improvements Needed: UX polish, deployment, documentation.

Result: AI functionality added securely using Supabase Edge Function.

Prompt Version: 6

Goal: Polish UX, validation, and deployment readiness.

Input Given: Existing working app and final MVP requirements.

Output Received: More polished MVP with validation, reusable components, README, and deployment preparation.

Problems Found: To be filled after final testing.

Result: Final working MVP prepared for deployment.

Prompt Engineering Techniques Used

To ensure high-quality, predictable outputs from the models, the following prompting strategies were used:

- Goal / Input / Output structure
- Role-based prompting
- Structured JSON responses
- Context injection
- Iterative refinement
- Error-driven prompt improvements

5. Project Iterations & Feature Evolution

The platform was built incrementally across 14 distinct development iterations:

- **Iteration 1:** Basic audit dashboard
- **Iteration 2:** Authentication
- **Iteration 3:** Project management
- **Iteration 4:** Risk assessment
- **Iteration 5:** Financial analyzer
- **Iteration 6:** Working paper generation

- **Iteration 7:** Audit procedures
- **Iteration 8:** Management letters
- **Iteration 9:** Planning memo generator
- **Iteration 10:** AI audit copilot
- **Iteration 11:** Knowledge hub
- **Iteration 12:** Prompt library
- **Iteration 13:** Immersive landing page
- **Iteration 14:** Dashboard improvements and premium features

6. Technical Review: What Worked vs. What Failed

What Worked Well

- Structured prompting
- Modular architecture
- Supabase integration
- Reusable React components
- JSON-based AI outputs

What Did Not Work Well & Challenges Faced

- Model compatibility issues
- Database schema mismatches and missing columns during feature expansion
- RLS policy configuration challenges
- Dashboard and active engagement synchronization issues

- Key system errors: Edge Function returned non-2xx status code , Gemini model not found , invokeGemini undefined , and schema cache errors (audit_scope , memo_text , priority columns, risk visualization data).

Problem-Solving Approach & Solutions Applied

Issues were resolved through:

1. **API Migration:** Migrated the core API infrastructure from Gemini to Groq to fix model availability and runtime errors.
2. **Database & Policy Updates:** Performed schema updates and corrected Supabase Row Level Security (RLS) policies.
3. **Prompt and Sync Refinements:** Improved prompt designs and implemented frontend/backend synchronization enhancements.

7. Final Product Features & Deliverables

Core Features

- User authentication
- Audit engagement and project management
- Dynamic dashboard and risk assessment engine
- Financial statement analyzer
- Audit checklist system

Premium Features

- AI-generated working papers
- AI-generated audit procedures
- AI-generated management letters

- AI-generated planning memos
- AI audit copilot
- Knowledge hub
- Prompt library

Final Result

The final result is a fully functional AI-powered audit platform that combines audit workflow management, risk assessment, documentation generation, and AI-assisted decision support into a single system.

8. Conclusion

AuditIQ Pro demonstrates how AI-assisted development can accelerate the creation of meaningful business applications. The project highlights the value of prompt engineering, iterative development, full-stack integration, and practical problem solving while delivering a real-world solution for the audit profession.